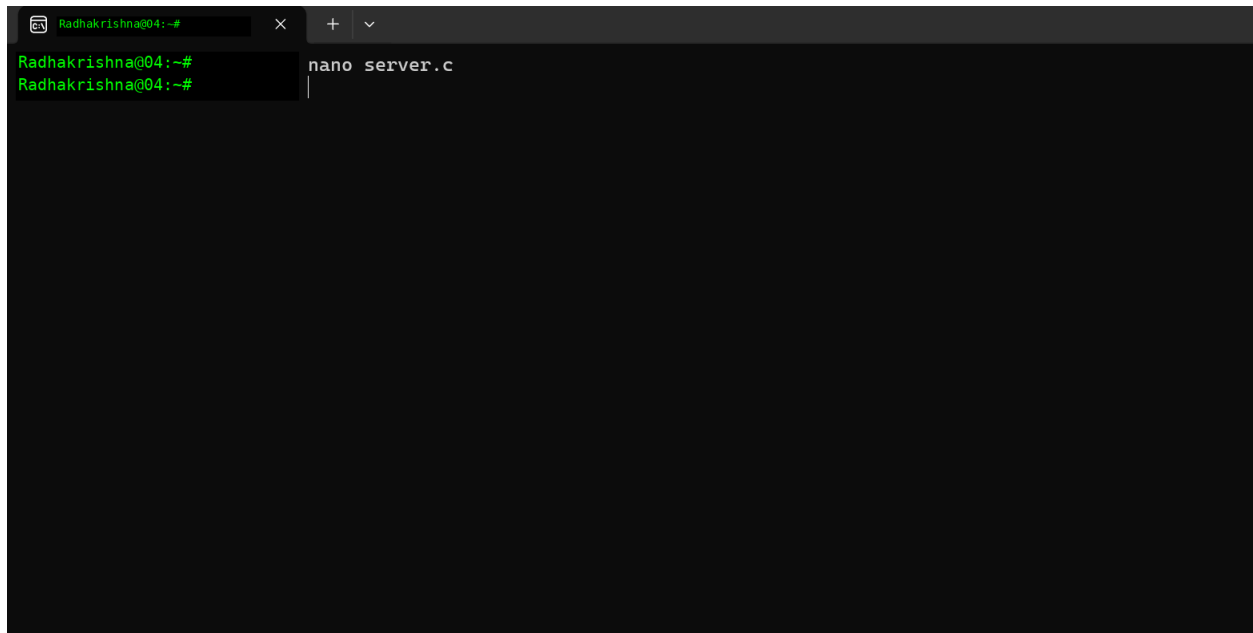


Practical-5

2520090063

Task-1

R.Radhakrishna

A screenshot of a terminal window. The title bar shows a tab with the text 'Radhakrishna@04: ~#'. The terminal content shows the prompt 'Radhakrishna@04:~#' followed by the command 'nano server.c'. The cursor is positioned at the end of the command line. The terminal background is dark, and the text is light green.

Code:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/stat.h>
#include <errno.h>

#define SERVER_FIFO "/tmp/server_fifo"
#define MAX_SIZE 512

int main() {
    int server_fd;
    char buffer[MAX_SIZE];
```

```
// Create server FIFO

if (mkfifo(SERVER_FIFO, 0666) == -1 && errno != EEXIST) {

    perror("mkfifo");

    exit(1);

}

printf("Server started...\n");

printf("Waiting for clients...\n");


// Open FIFO for reading

server_fd = open(SERVER_FIFO, O_RDONLY);

if (server_fd == -1) {

    perror("open");

    exit(1);

}

while (1) {

    memset(buffer, 0, sizeof(buffer));

    int bytes = read(server_fd, buffer, sizeof(buffer) - 1);

    if (bytes > 0) {

        buffer[bytes] = '\0';

        // Message format: clientPID:message

        char *colon = strchr(buffer, ':');

        if (colon != NULL) {

            *colon = '\0';

            char *client_pid = buffer;

            char *message = colon + 1;
```

```

printf("\nReceived from Client %s: %s\n",
      client_pid, message);

// Create response FIFO for this client

char response_fifo[100];

snprintf(response_fifo, sizeof(response_fifo),
        "/tmp/client_%s_fifo", client_pid);

int response_fd = open(response_fifo, O_WRONLY);

if (response_fd != -1) {
    char response[MAX_SIZE];

```

output:

```

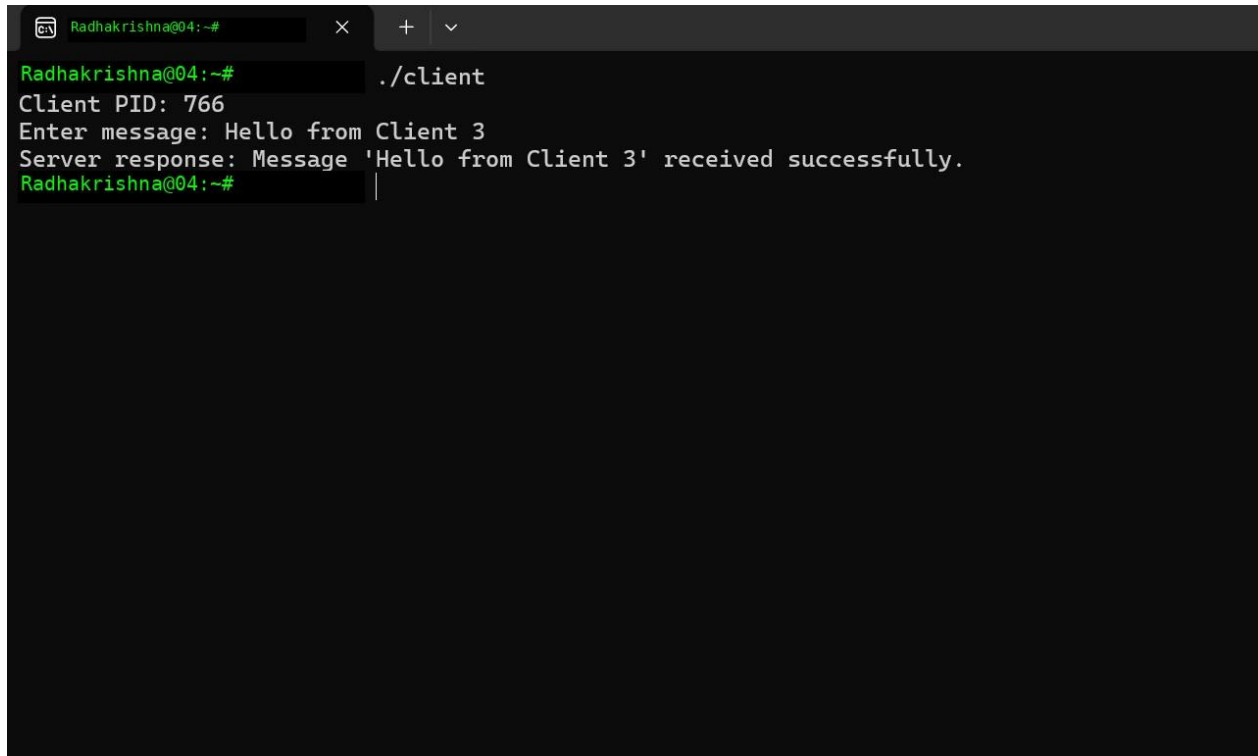
Radhakrishna@04:~# nano server.c
Radhakrishna@04:~# nano server.c
Radhakrishna@04:~# gcc server.c -o server
server.c: In function 'main':
server.c:55:39: warning: '%s' directive output may be truncated
writing up to 511 bytes into a region of size 88 [-Wformat-trunc
ation=]
55 |         "/tmp/client_%s_fifo", client_p
id);
|
server.c:54:17: note: 'snprintf' output between 18 and 529 bytes
into a destination of size 100
54 |         snprintf(response_fifo, sizeof(response_
fifo),
|
55 |         "/tmp/client_%s_fifo", client_p
id);
|
Radhakrishna@04:~# ls
PID.c      Sum.c      linux.c    os.c       prog.c     s
kill4task1.c skill4task3.c skill4task5.c task03.c
PIDs.c     client     myfile.txt process     server     s
kill4task2.c skill4task4.c task01     task1.c
PIDs.c.save client.c     notes.txt.save process.c   server.c   s
kill4task2.c skill4task4.c task01.c   task1.c.save
Sum         FIRST.C    os         prog        skill4task1 s
kill4task3  skill4task5 task03
Radhakrishna@04:~# ./server
Server started...
Waiting for clients...

Received from Client 699: Hello Server

Radhakrishna@04:~# nano client.c
Radhakrishna@04:~# gcc client.c -o client
client.c: In function 'main':
client.c:46:18: warning: '%s' directive output may be truncated
writing up to 511 bytes into a region of size between 500 and 51
0 [-Wformat-truncation=]
46 |         "%d:%s", pid, message);
|
client.c:45:5: note: 'snprintf' output between 3 and 524 bytes i
nto a destination of size 512
45 |         snprintf(buffer, sizeof(buffer),
|
46 |         "%d:%s", pid, message);
|
Radhakrishna@04:~# ls
PID.c      Sum.c      linux.c    os.c       prog.c     s
kill4task1.c skill4task3.c skill4task5.c task03.c
PIDs.c     client     myfile.txt process     server     s
kill4task2.c skill4task4.c task01     task1.c
PIDs.c.save client.c     notes.txt.save process.c   server.c   s
kill4task2.c skill4task4.c task01.c   task1.c.save
Sum         FIRST.C    os         prog        skill4task1 s
kill4task3  skill4task5 task03
Radhakrishna@04:~# ./client
Client PID: 699
Enter message: Hello Server
Server response: Message 'Hello Server' received successfully.
Radhakrishna@04:~#

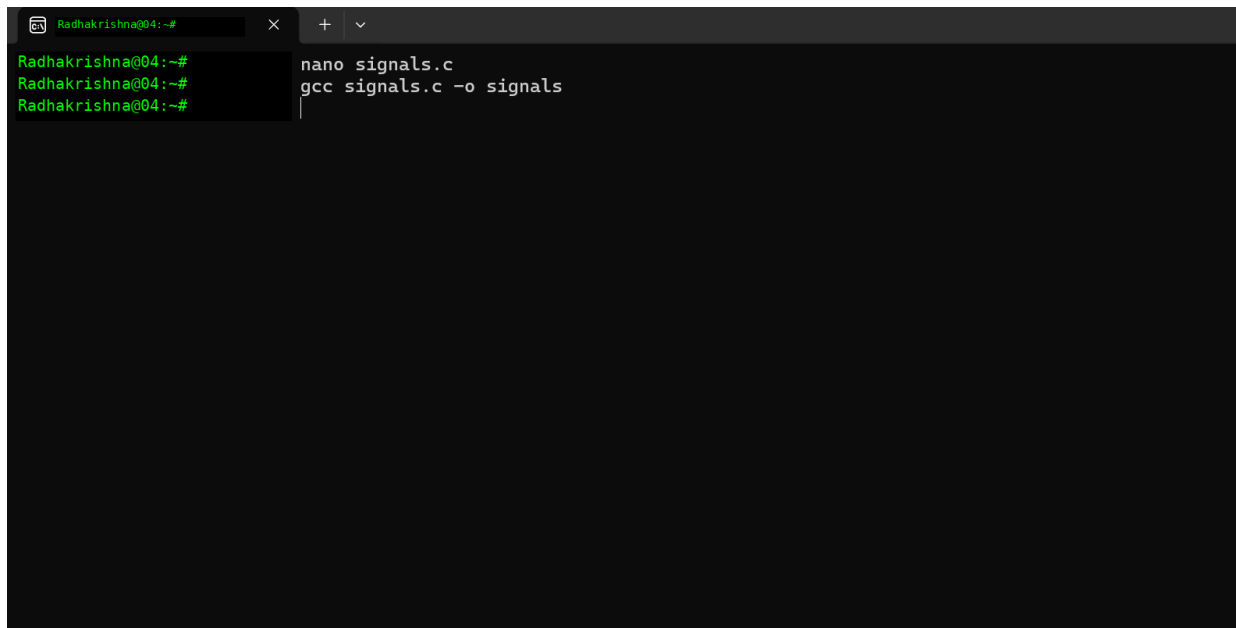
```

Multiple terminals:

A terminal window titled 'Radhakrishna@04:~#' with a dark background and green text. The prompt 'Radhakrishna@04:~#' is followed by the command './client'. The output shows 'Client PID: 766', a prompt 'Enter message: Hello from Client 3', and a response 'Server response: Message 'Hello from Client 3' received successfully.'.

```
Radhakrishna@04:~# ./client
Client PID: 766
Enter message: Hello from Client 3
Server response: Message 'Hello from Client 3' received successfully.
Radhakrishna@04:~#
```

Task-2:

A terminal window titled 'Radhakrishna@04:~#' with a dark background and green text. The prompt 'Radhakrishna@04:~#' is followed by the command 'nano signals.c'. The prompt is repeated, followed by 'gcc signals.c -o signals'. The prompt is repeated again.

```
Radhakrishna@04:~# nano signals.c
Radhakrishna@04:~# gcc signals.c -o signals
Radhakrishna@04:~#
```

```
Radhakrishna@04:~# nano signals.c
Radhakrishna@04:~# gcc signals.c -o signals
Radhakrishna@04:~# ./signals
Signal handling program started.
Process ID: 808
Waiting for signals...

Radhakrishna@04:~# kill -SIGUSR1 2456
-bash: kill: (2456) - No such process
Radhakrishna@04:~#
```

```
SIGTERM received!
Termination request received.
Program terminating...
Program ended.
```